USN | | | | | | | | | | | [23TCSE311]

THIRD SEMESTER B.Tech EXAMINATION, NOVEMBER-2024

SCRIPTING FOR HACKERS

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A			**5 x2=10**
1.	**a.**	Mention the functions of "import" statement in python	Remember	CO1	**02**
	b.	Define Wireshark.	Remember	CO2	**02**
	c.	Write a short note on brute force attack.	Remember	CO3	**02**
	d.	Define transposition cipher.	Understand	CO4	**02**
	e.	Mention the purpose of Mac address.	Remember	CO4	**02**
		PART-B			**6 X 5=30**
2.	**a.**	Explain the various types of Hacking techniques in networks.	Understand	CO1	**05**
		OR			
	b.	Explain the TCP/IP model and its protocol in detail.	Understand	CO1	**05**
3.	**a.**	Give the detail about ICMP? Explain its working.	Understand	CO2	**05**
		OR			
	b.	What is ARP cache poisoning? Explain how it affects the network functioning.	Understand	CO2	**05**
4.	**a.**	Consider an IP address as sender address. Write a Python script to find the location of this sender using ipinfo.io API. Explain its working.	Apply	CO3	**05**
		OR			
	b.	You are assigned to assess the security of a wireless network at your company. Using Scapy in Python, write a script that can detect nearby Wi-Fi networks by passively listening to Wi-Fi beacon frames.	Apply	CO3	**05**
5.	**a.**	Write a python code for applying brute force attack in reverse cipher.	Apply	CO4	**05**
		OR			
	b.	Demonstrate the transposing cipher with an example.	Apply	CO4	**05**
6.	**a.**	Discuss how TOR browser supports the illegal activities.	Apply	CO5	**05**
		OR			
	b.	Explain the working of MAC address in detail.	Understand	CO5	**05**
7.	**a.**	Explain about the various tools used for packet sniffing on Windows.	Understand	CO6	**05**
		OR			

	b.	Explain MiTM and Email credential stealing attack in detail.	Understand	CO6	**05**
		PART-C			**6 X 10=60**
8.	**a.**	Write a python program to scan the port with the intention of detecting the port. Explain the concept in detail.	Apply	CO2	**10**
		OR			
	b.	Build a python module name "myHack.py" and use it to set up a hacking environment using Kali Linux.	Apply	CO2	**10**
9.	**a.**	Demonstrate ICMP Echo request and reply message capture in wireshark. Explain how to use it.	Apply	CO3	**10**
		OR			
	b.	Illustrate how to capture the TCP packets in port 80. Analyze the captured packet.	Apply	CO3	**10**
10.	**a.**	Consider a server with the address 192.168.1.124. Demonstrate how to automate the collecting packets from FTP, HTTP and SMTP. Construct a program to collect data from these ports and demonstrate how to be analyzed for security vulnerabilities.	Apply	CO4	**10**
		OR			
	b.	Write a Python program to get the geolocation data of the IP and explain how geolocation can assist in cyber investigations. Use ipapi.io to collect the geo-location	Apply	CO4	**10**
11.	**a.**	Explain ceaser cipher and write a program to encrypt and decrypt the input "Hi How are You" using ceaser cipher.	Apply	CO6	**10**
		OR			
	b.	Write the program for Brute-Force Approach- reverse cipher. Explain the concepts in detail.	Apply	CO6	**10**
12.	**a.**	Explain Virtual private network. Write a python program to create a VPN.	Apply	CO5	**10**
		OR			
	b.	Explain the dark web and demonstrate how to approach it safely.	Apply	CO5	**10**
13.	**a.**	Give few third party python libraries that supports for hacking. Explain its purpose in detail.	Understand	CO1	**10**
		OR			
	b.	Sketch the structure of an IP packet. Explain the functions of each parameters of the packet.	Understand	CO1	**10**